WORKED SOLUTION

STEP I 2006, Question 5

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q5

The key idea

Both integrands contain an awkward square root, and the cleanest way to kill a square root of a linear expression is to call the whole root a new variable. Once the root has gone, what is left is a rational function, which partial fractions can always integrate. In part (ii) the exponentials are best removed first, so that the square root becomes the root of something linear, and then the same trick applies.

Solution

Part (i)

Since $x > 4$, we have $2x + 1 > 9$, so we may take $u = \sqrt{2x+1} > 3$. From $u^2 = 2x + 1$,

$$2u\,\mathrm{d}u = 2\,\mathrm{d}x, \qquad \text{so} \qquad \mathrm{d}x = u\,\mathrm{d}u,$$

and

$$x - 4 = \frac{u^2 - 1}{2} - 4 = \frac{u^2 - 9}{2}.$$

Substituting,

$$\int \frac{3}{(x-4)\sqrt{2x+1}}\,\mathrm{d}x = \int \frac{3}{\frac{1}{2}(u^2-9)\,u}\,u\,\mathrm{d}u = \int \frac{6}{u^2-9}\,\mathrm{d}u.$$

The factor u from $\mathrm{d}x$ cancels the square root exactly; that is why this substitution works. Now $u^2 - 9 = (u-3)(u+3)$, and

$$\frac{6}{(u-3)(u+3)} = \frac{1}{u-3} - \frac{1}{u+3},$$

as you can check by recombining: the numerator is $(u+3) - (u-3) = 6$. Hence

$$\int \frac{6}{u^2-9}\,\mathrm{d}u = \ln(u-3) - \ln(u+3) + K = \ln\left(\frac{u-3}{u+3}\right) + K,$$

where no modulus signs are needed because $u > 3$ makes both $u-3$ and $u+3$ positive. Putting $u = \sqrt{2x+1}$ back gives

$$\int \frac{3}{(x-4)\sqrt{2x+1}}\,\mathrm{d}x = \ln\left(\frac{\sqrt{2x+1}-3}{\sqrt{2x+1}+3}\right) + K,$$

as required.

Part (ii)

Let I be the integral. The obvious substitution, copying part (i), is to name the square root:

$$w = \sqrt{\mathrm{e}^x + 1}, \qquad w^2 = \mathrm{e}^x + 1.$$

Then $2w\,\mathrm{d}w = \mathrm{e}^x\,\mathrm{d}x$, so $\mathrm{d}x = \dfrac{2w}{\mathrm{e}^x}\,\mathrm{d}w = \dfrac{2w}{w^2-1}\,\mathrm{d}w$. The limits become

$$x = \ln 3 :\ w = \sqrt{3+1} = 2, \qquad x = \ln 8 :\ w = \sqrt{8+1} = 3.$$

Therefore

$$I = \int_2^3 \frac{2}{(w^2-1)\,w} \cdot \frac{2w}{w^2-1}\,\mathrm{d}w = \int_2^3 \frac{4}{(w^2-1)^2}\,\mathrm{d}w.$$

Ask yourself How do we split $\dfrac{4}{(w^2-1)^2}$ into partial fractions without a long calculation?

Start from the simple split $\dfrac{2}{w^2-1} = \dfrac{1}{w-1} - \dfrac{1}{w+1}$ and square it:

$$\begin{aligned}\frac{4}{(w^2-1)^2} &= \frac{1}{(w-1)^2} - \frac{2}{(w-1)(w+1)} + \frac{1}{(w+1)^2} \\ &= \frac{1}{(w-1)^2} + \frac{1}{(w+1)^2} - \frac{1}{w-1} + \frac{1}{w+1},\end{aligned}$$

using the simple split once more on the middle term. Each piece now integrates at sight:

$$\int \frac{4}{(w^2-1)^2}\,\mathrm{d}w = -\frac{1}{w-1} - \frac{1}{w+1} - \ln\left(\frac{w-1}{w+1}\right) + C,$$

with no modulus signs needed since $w \geq 2$ on the range of integration. Evaluating,

$$\begin{aligned}\text{at } w = 3 :&\quad -\tfrac{1}{2} - \tfrac{1}{4} - \ln\tfrac{2}{4} = -\tfrac{3}{4} + \ln 2, \\ \text{at } w = 2 :&\quad -1 - \tfrac{1}{3} - \ln\tfrac{1}{3} = -\tfrac{4}{3} + \ln 3.\end{aligned}$$

So

$$I = \left(-\tfrac{3}{4} + \ln 2\right) - \left(-\tfrac{4}{3} + \ln 3\right) = \frac{7}{12} + \ln\frac{2}{3},$$

which is the result we had to show.

You could equally do this in two steps, first $v = \mathrm{e}^x$ (giving $\int_3^8 \frac{2}{v^2\sqrt{v+1}}\,\mathrm{d}v$) and then $w^2 = v+1$; the single substitution above does both at once.

Checks

- Differentiate the answer to (i): with $u = \sqrt{2x+1}$, $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}$, and $\frac{\mathrm{d}}{\mathrm{d}u}\ln\frac{u-3}{u+3} = \frac{6}{u^2-9}$, so the derivative is $\frac{6}{u(u^2-9)} = \frac{3}{(x-4)\sqrt{2x+1}}$.
- Numerically, $\frac{7}{12} + \ln\frac{2}{3} \approx 0.58333 - 0.40547 = 0.17787$, and Simpson's rule applied to the original integrand on $[\ln 3, \ln 8]$ gives 0.177868, agreeing to six figures.
- A sanity check on sign and size: the integrand in (ii) is positive and falls from about 0.33 to about 0.08 over an interval of length $\ln\frac{8}{3} \approx 0.98$, so an answer a little under 0.2 is just what we expect.

Common slips

- Forgetting to change the limits in (ii) when changing variable, or substituting x-limits into a w-expression.
- Writing $\frac{4}{(w^2-1)^2}$ as only $\frac{A}{(w-1)^2} + \frac{B}{(w+1)^2}$: the repeated factors need the linear terms $\frac{1}{w-1}$ and $\frac{1}{w+1}$ as well.
- Getting $\mathrm{d}x = \mathrm{d}u$ instead of $\mathrm{d}x = u\,\mathrm{d}u$ in (i), which leaves a stray square root.

Beyond STEP

The reason both parts end in logarithms and rational terms is a general theorem: every rational function has an elementary antiderivative built from rational functions, logarithms and inverse tangents, via partial fractions; and any integrand that is rational in x and $\sqrt{ax+b}$ becomes rational after the substitution $u^2 = ax + b$. Extending this to roots of quadratics leads to Euler's substitutions, and asking which integrals *cannot* be done in elementary terms leads to Liouville's theory of integration in finite terms. For a careful treatment of integration and why it works, read Abbott, *Understanding Analysis*.

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